Database Normalisation

Normalisation was considered when designing the FinTrust database to reduce repeated information and make the records easier to manage.

The database was organised into six related tables instead of storing all the information in one large table.

First Normal Form (1NF)

The database satisfies First Normal Form because each field stores a single value, and each record can be identified individually.

For example, the customers table stores the first name, last name, email and phone number in separate columns. Each customer has a unique customer ID. Transaction records are also stored separately rather than grouping multiple transactions in one record.

Second Normal Form (2NF)

The database satisfies Second Normal Form because all the tables have single-column primary keys. This means there are no partial dependencies where an attribute depends on only part of a composite primary key.

For example, customer details depend on the customer ID, while account details depend on the account ID.

Third Normal Form (3NF)

The database was also organised to meet Third Normal Form by separating information that belongs to different entities.

For example, transaction types and transaction statuses are stored in their own tables. The transactions table only stores their IDs as foreign keys instead of repeating the type and status names in every transaction record.

Customer information is also kept separately from account information, with the customer ID linking each account to its owner.

This design reduces unnecessary repetition and makes it easier to update customer details, transaction types and transaction statuses without changing multiple transaction records.

Conclusion

The use of normalisation helped organise the FinTrust database into related tables, reduce data duplication and maintain consistency across records.